Bayesian Risk Analysis of Perception-Based Controllers via RoVer-CoRe

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***Abstract*—Autonomous systems frequently rely on high-dimensional perception modules that introduce significant uncertainty into the control loop. While formal verification techniques, such as Hamilton-Jacobi (HJ) reachability, provide rigorous worst-case safety guarantees [2], [3], they are often excessively conservative. By treating all admissible perception errors as equally likely within a bounded uncertainty set, these methods frequently classify functional states as "unsafe," even when the probabilistic risk of failure is negligible under realistic conditions.**

This paper presents a Bayesian risk analysis framework designed to quantify the conservatism gap in perception-based controllers. Utilizing the RoVer-CoRe verification environment [1], we model perception errors as stochastic processes governed by parametric distributions rather than static bounded sets. We employ Markov Chain Monte Carlo (MCMC) sampling via the No-U-Turn Sampler (NUTS) [6] to characterize systematic biases and variance in state estimation across two rover navigation benchmarks, including 3-D and 4-D simulations. Our approach bridges the gap between formal worst-case guarantees and probabilistic operational reality by replaying recorded control sequences under posterior-predictive noise distributions, enabling a nuanced mapping of failure risk across the initial state space. Our results reveal a quantifiable conservatism gap in which a substantial portion of formally "unsafe" states exhibit posterior predictive failure probabilities well below 50%, suggesting that worst-case verification alone is an incomplete picture of operational safety.

***Index Terms*—Bayesian inference, Hamilton-Jacobi reachability, perception-based control, autonomous systems, MCMC, safety verification, posterior predictive, RoVer-CoRe**

I. Introduction

As autonomous systems transition from controlled laboratory environments to unpredictable real-world scenarios, ensuring their safety remains a paramount engineering challenge. Modern rovers, unmanned aerial vehicles, and autonomous ground vehicles typically employ perception-based controllers in which state estimates are derived from high-dimensional sensor data using learned models [1]. These perception modules introduce inherent uncertainties ranging from zero-mean Gaussian noise to systematic biases induced by training distribution shift [7] that can propagate through the control loop and lead to catastrophic failures.

Formal verification methods, specifically Hamilton-Jacobi (HJ) reachability analysis, have emerged as a principled tool for providing rigorous safety guarantees [2], [3]. By computing a Backward Reachable Tube (BRT), these methods identify the set of all initial states from which a system might enter a failure set under worst-case realization of bounded uncertainty [4]. While mathematically rigorous, this worst-case paradigm often results in extreme conservatism [5]. In practice, many states labeled as unsafe by a BRT may fail only under highly improbable noise sequences, leading to operational restrictions that unnecessarily prevent autonomous agents from using viable regions of the state space.

This paper addresses this conservatism gap by introducing a Bayesian risk analysis framework built upon the RoVer-CoRe verification environment [1]. Rather than asking whether failure is *possible* under some admissible error, we ask how *likely* failure is under a probabilistic model of perception error learned directly from simulation data. Our contributions are threefold:

- We employ MCMC via NUTS [6] to characterize the stochastic nature of perception errors in the RoVer-CoRe benchmarks, moving beyond bounded sets to full posterior distributions that capture systematic biases and aleatoric variance.
- We implement a replay-based posterior predictive simulation that couples these learned distributions with a data-driven approximation of the system's feedback controller, accounting for the time-varying growth of perception uncertainty observed in the RoVer-CoRe environment.
- We quantify the conservatism gap by mapping posterior predictive failure probabilities across a dense grid of 135 initial states, revealing the practical divergence between binary HJ reachability classifications and probabilistic operational risk.

The remainder of this paper is organized as follows. Section II reviews the relevant background and related work. Section III describes our data collection, Bayesian model, and simulation methodology. Section IV presents the empirical results. Section V analyzes the conservatism gap and the impact of systematic bias. Section VI concludes with limitations and directions for future work.

II. Background and Related Work

A. Hamilton-Jacobi Reachability and the BRT

The safety of a continuous-time dynamical system $\dot{x} = f(x, u, d)$ with state $x \in \mathbb{R}^n$, control $u \in \mathcal{U}$, and disturbance

$d \in \mathcal{D}$, is canonically analyzed through a payoff function $l(x)$, where $l(x) \leq 0$ denotes membership in the failure set $\mathcal{F}$ [2]. HJ reachability computes a value function $V(x,t)$ representing the minimum payoff achievable by an adversarial disturbance over the time horizon $[t,T]$:

$$V(x,t) = \min_{d(\cdot)\in\mathcal{D}} l(x(T)) \tag{1}$$

The BRT at time $t = 0$ is then the zero-sublevel set of this value function:

$$\mathcal{R} = \{x : V(x,0) < 0\} \tag{2}$$

Any initial state $x_0 \in \mathcal{R}$ is guaranteed to enter the failure set under *some* admissible disturbance trajectory, while any $x_0 \notin \mathcal{R}$ is guaranteed safe under all admissible disturbances [3]. Mitchell et al. [3] established the theoretical foundation for computing these sets via the viscosity solution of a Hamilton-Jacobi partial differential equation, which Bansal et al. [2] subsequently extended with efficient GPU-parallelized numerical tools.

In the perception-based control setting formalized by RoVer-CoRe [1], the disturbance d encodes the perception error $e = \hat{x} - x$, where $\hat{x}$ is the estimated state fed to the controller. Standard RoVer-CoRe implementations assume e lies within a compact set $\mathcal{E}$, typically derived from empirical error bounds at a specified coverage level.

B. Stochastic and Probabilistic Reachability

The conservatism of worst-case BRT analysis has motivated a parallel literature on stochastic and probabilistic reachability. Summers and Lygeros [5] developed a dynamic programming framework that computes the probability of a discrete-time stochastic system remaining within a safe set over a finite horizon, yielding probabilistic rather than binary safety certificates. While theoretically grounded, these methods are often computationally intractable for the nonlinear, continuous-time systems found in modern autonomous navigation, and they require a fully specified prior distribution over disturbances.

Our approach differs in that we *learn* the distribution of perception error from simulation data via Bayesian inference, rather than specifying it a priori. We then use Monte Carlo simulation to estimate failure probabilities under the learned distribution, providing a data-driven complement to the existing theoretical frameworks.

C. Conformal Prediction and Safe Learning

A complementary line of work applies conformal prediction to provide finite-sample coverage guarantees on uncertainty sets for safe learning-based systems [7]. These methods guarantee that a prediction set contains the true label with at least a desired probability, without making distributional assumptions. Luo et al. [8] combine conformal prediction with dynamics simulators to construct provably safe warning systems requiring as few as $1/\varepsilon$ calibration examples. While conformal methods offer distribution-free guarantees, they operate on the uncertainty *set* rather than the full distribution. Our Bayesian approach instead provides the full posterior over risk, including credible intervals and posterior predictive summaries, which are more informative for decision-making under uncertainty.

D. RoVer-CoRe

Lin, Pinto, and Bansal [1] introduced RoVer-CoRe as the first HJ-reachability-based framework for formal verification of perception-based systems under perceptual uncertainty. The key insight is to concatenate the system controller, observation function, and state estimation modules into a single equivalent closed-loop system amenable to standard HJ analysis. RoVer-CoRe demonstrates strong empirical results on aircraft taxiing and neural-network-based rover navigation benchmarks, including the RoverDark and RoverLight case studies that constitute the dataset for this work. The framework explicitly models perception uncertainty as a bounded set $\mathcal{E}$ and identifies future directions including probabilistic uncertainty models [1] — the extension this paper begins to address.

III. Methods

Our methodology decomposes into three phases: (1) data ingestion and error pre-processing, (2) Bayesian parameter estimation via MCMC, and (3) a coupled-control replay simulation for posterior predictive failure estimation.

A. Data Source and Collection

The dataset originates from controlled simulation experiments conducted using the open-source RoVer-CoRe environment [1]. This is experimental, not observational, data: each run fixes a controller and initial condition, simulates the closed-loop system forward under prescribed dynamics, and records the full state and perception estimate trajectories. No passive observation of real-world systems is involved.

We utilize two benchmark case studies:

- **RoverDark**: A Dubins-car rover navigating an obstacle field in low-light conditions under a Model Predictive Control (MPC) policy, with 3-dimensional state (p_x, p_y, θ) representing forward position, lateral position, and heading angle.
- **RoverLight**: The same rover equipped with a BRT-scheduled light activation policy that reduces perception uncertainty when the rover approaches the boundary of the BRT, with 4-dimensional state (p_x, p_y, θ, s) where s indicates the light activation state.

Each case study contributes 6 simulated trajectories of 500 timesteps at $\Delta t = 0.01\,\text{s}$, yielding a total horizon of $T = 5\,\text{s}$. Pooled across both benchmarks, we obtain 6,000 perception error samples per state dimension.

B. Data Pre-Processing and Exploratory Analysis

For each trajectory i and timestep t, we compute the perception error vector:

$$e_t = \hat{x}_t - x_t \in \mathbb{R}^3 \tag{3}$$

where we retain only the (p_x, p_y, θ) components. The light-state dimension s of the RoverLight trajectories carries no perception error (verified numerically to be identically zero) and is discarded.

A critical pre-processing step is the identification and removal of *structural zeros*: error samples that are exactly zero in all three dimensions simultaneously. These arise at $t = 0$ (before the uncertainty model has grown from its initial value) and after light-activation reset events in RoverLight (where the BRT-scheduled policy resets the estimator state, momentarily zeroing the error). These samples are not realizations of the perception noise process; including them biases the sample mean toward zero and artificially deflates the estimated variance. After removing structural zeros, we retain $N_{\text{eff}} = 5{,}612$ error samples for model fitting.

Exploratory analysis reveals three key features of the error distribution:

1) **Systematic bias**: All three error components have statistically credible nonzero means: $\bar{e}_{p_x} \approx -0.066\,\text{m}$, $\bar{e}_{p_y} \approx -0.028\,\text{m}$, and $\bar{e}_\theta \approx -0.008\,\text{rad}$, suggesting consistent leftward and downward shifts attributable to training distribution shift.
2) **Time-varying uncertainty**: The standard deviation of the lateral error σ_{p_y} grows approximately threefold over the 5-second horizon, consistent with RoVer-CoRe's linear uncertainty growth model $\varepsilon(t) = \varepsilon_0 + kt$ [1].
3) **Approximate Gaussianity**: The nonzero error samples are approximately symmetric about their means and unimodal, supporting a Normal likelihood as a reasonable starting model.

C. Bayesian Model Specification

We adopt a conditionally independent Gaussian model for the perception error, treating each state dimension $d \in \{p_x, p_y, \theta\}$ separately. Given N nonzero error samples $\{e_i^{(d)}\}_{i=1}^N$, the model is:

$$e_i^{(d)} \mid \mu_d, \sigma_d \sim \mathcal{N}(\mu_d, \sigma_d^2) \tag{4}$$

with weakly informative priors reflecting our prior beliefs about perception error magnitudes:

$$\mu_d \sim \mathcal{N}(0,\, 0.5^2), \qquad \sigma_d \sim \text{HalfNormal}(0.3) \tag{5}$$

The prior on μ_d is centered at zero because estimators are designed to minimize expected error, making systematic bias unlikely a priori but not excluded. The HalfNormal(0.3) prior on σ_d is consistent with the empirical spread of 0.1–0.2 m for position errors and ≈0.03 rad for heading errors reported in the RoVer-CoRe benchmarks [1], while remaining broad enough to accommodate larger uncertainties.

The joint posterior over all parameters factors as:

$$p(\boldsymbol{\mu}, \boldsymbol{\sigma} \mid \mathcal{D}) \propto \prod_d p(\mu_d) p(\sigma_d) \prod_{i=1}^N \mathcal{N}(e_i^{(d)} \mid \mu_d, \sigma_d^2) \tag{6}$$

Because this model has a conditionally conjugate structure (Normal-InvGamma with known hyperparameters), the posterior admits a closed-form analytic solution as a verification check. We confirmed that the analytic posterior mean and variance match the MCMC estimates to within numerical precision.

D. Posterior Inference via MCMC

We perform posterior inference using the No-U-Turn Sampler (NUTS) [6] implemented in PyMC. NUTS is an adaptive variant of Hamiltonian Monte Carlo (HMC) that automatically tunes the number of leapfrog integration steps, eliminating the need for user-specified trajectory length and enabling efficient exploration of posterior geometry in moderate-dimensional spaces. We run 2 chains with 1,000 tuning steps and 2,000 posterior draws per chain, collecting 4,000 total posterior samples.

Convergence is assessed via the Gelman-Rubin potential scale reduction factor $\hat{R}$ and the effective sample size (ESS). For all parameters, $\hat{R} \leq 1.002$ and bulk ESS $> 4{,}900$, indicating thorough mixing and convergence to the target posterior.

E. Posterior Predictive Failure Probability

The central quantity of interest is the posterior predictive failure probability for a given initial state x_0:

$$P(\text{fail} \mid x_0) = \mathbb{E}_{(\boldsymbol{\mu},\boldsymbol{\sigma}) \sim p(\cdot|\mathcal{D})} \left[P(\text{fail} \mid x_0, \boldsymbol{\mu}, \boldsymbol{\sigma}) \right] \tag{7}$$

The inner expectation is intractable in closed form because it requires propagating Gaussian noise through nonlinear closed-loop dynamics. We therefore estimate it via Monte Carlo simulation: for each of $K = 150$ posterior draws $(\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)$, we simulate $M = 100$ closed-loop trajectories from x_0 and record the fraction that fail. The posterior predictive mean is the average over draws:

$$\widehat{P}(\text{fail} \mid x_0) = \frac{1}{K} \sum_{k=1}^K \hat{p}_k(x_0) \tag{8}$$

where $\hat{p}_k(x_0) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[\text{fail}_{k,m}]$ is the empirical failure rate under posterior draw k. The variance across draws $\hat{p}_k(x_0)$ yields 95% credible intervals for the failure probability.

F. Coupled-Control Replay Simulation

The simulation environment requires a closed-loop controller approximating the MPC behavior recorded in the RoVer-CoRe trajectories. We fit a linear proportional controller via Ordinary Least Squares (OLS) regression on the recorded $(u_t, \hat{x}_t)$ pairs:

$$u_t \approx -k_{p_y} \hat{p}_{y,t} - k_\theta \hat{\theta}_t \tag{9}$$

yielding fitted gains k_{p_y} and k_θ. We note that a linear model captures only a small fraction of the MPC's behavioral variance ($R^2 = 0.017$), as the true controller is nonlinear and state-history-dependent. This is acknowledged as the primary approximation in our methodology; see Section VI.

To capture the non-stationary nature of the uncertainty model, we incorporate a time-varying noise scaling factor derived from the empirical RoverDark error statistics. At each timestep t, the posterior standard deviation is scaled by the observed empirical growth ratio:

$$\sigma_t^{(d)} = \sigma_{\text{posterior}}^{(d)} \cdot \frac{\sigma_{\text{empirical}}^{(d)}(t)}{\sigma_{\text{empirical}}^{(d)}(0) + \epsilon} \tag{10}$$

where $\epsilon = 10^{-8}$ prevents division by zero at $t = 0$. This scaling ensures that early-trajectory errors are small (matching the initial uncertainty bound ε_0) and late-trajectory errors are larger (matching the grown bound $\varepsilon_0 + kT$).

The rover dynamics in each rollout follow a Dubins-car model with Euler integration:

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ u \end{bmatrix} \tag{11}$$

with forward speed $v = 1.5\,\text{m/s}$ and $\Delta t = 0.01\,\text{s}$. Failure is declared if $|p_y(t)| \geq 1.5\,\text{m}$ at any step — a heuristic lateral boundary consistent with the navigable corridor in the RoVer-CoRe benchmarks. We evaluate this procedure across a grid of 135 initial states sweeping $p_y \in [-1.4, 1.4]\,\text{m}$ and $\theta \in [-0.8, 0.8]\,\text{rad}$ at fixed $p_x = 0\,\text{m}$, yielding approximately 1.35×10^7 total trajectory evaluations.

IV. Results

A. Posterior Characterization of Perception Error

Table I summarizes the posterior distributions for all six parameters. The posteriors are remarkably tight: with $N = 5{,}612$ samples and a three-parameter Gaussian model per dimension, the posterior uncertainty about σ is on the order of 0.001–0.003. This is expected given the large sample size relative to model complexity.

TABLE I
Posterior Summary for Perception Error Parameters (NUTS, 4,000 draws)

Param.	**Mean**	**SD**	**95% HDI**	$\hat{R}$
μ_{p_x}	-0.06576	0.00193	$[-0.069, -0.062]$	1.000
μ_{p_y}	-0.02812	0.00220	$[-0.032, -0.024]$	1.001
μ_θ	-0.00793	0.00037	$[-0.009, -0.007]$	1.001
σ_{p_x}	0.15170	0.00140	$[0.149, 0.154]$	1.000
σ_{p_y}	0.17269	0.00156	$[0.170, 0.176]$	1.000
σ_θ	0.02857	0.00026	$[0.028, 0.029]$	1.002

The key finding is that all three posterior means for μ are credibly nonzero: none of the 95% highest density intervals (HDIs) contains zero. This confirms a statistically significant systematic bias in the perception system across all dimensions. The posterior mean for σ_{p_y} ($0.1727\,\text{m}$) is nearly an order of magnitude larger than for σ_θ ($0.0286\,\text{rad}$), reflecting that lateral position estimation is substantially noisier than heading estimation in this benchmark.

B. Posterior Predictive Check

Fig. 3 presents the posterior predictive check, overlaying the empirical distribution of nonzero error samples against draws from the posterior predictive distribution. The Gaussian model reproduces the scale and approximate shape of the continuous error component well across all three dimensions. The posterior predictive distribution for p_y and θ is visually indistinguishable from the empirical distribution, and the p_x dimension shows only minor discrepancy in the tails.

The check reveals, however, that the raw data contains a non-trivial fraction of structural zeros (estimated at 6.5% of total samples), which the Gaussian model cannot reproduce by construction. As discussed in Section III, these zeros arise from initialization at $t = 0$ and from light-reset events in RoverLight. This constitutes a genuine model misspecification in the raw data but does not affect the fit on the nonzero component used for inference.

C. Posterior Predictive Failure Risk Grid

Table II presents a representative subset of the posterior mean failure probabilities $\widehat{P}(\text{fail} \mid x_0)$ over the (p_y, θ) grid, and Fig. 1 shows the full heatmap.

TABLE II
Posterior Mean Failure Probabilities $\widehat{P}(\text{fail} \mid x_0)$ with 95% Credible Intervals at Selected States

p_y	$\theta = -46°$	$\theta = -23°$	$\theta = 0°$	$\theta = +23°$	$\theta = +46°$
$+1.40\,\text{m}$	0.941	0.549	0.640	0.956	1.000
$+0.60\,\text{m}$	0.977	0.695	0.453	0.669	0.987
$0.00\,\text{m}$	0.994	0.793	0.421	0.559	0.994
$-0.60\,\text{m}$	1.000	0.896	0.455	0.472	0.981
$-1.20\,\text{m}$	1.000	0.988	0.558	0.425	0.948

Several structural features of the risk landscape are immediately apparent:

1) **Heading dominates lateral position.** Even at $p_y = 0$ (the center of the corridor), $\widehat{P}(\text{fail}) \approx 0.99$ at large heading angles ($|\theta| \approx 46°$). By contrast, states with $|p_y| = 1.2\,\text{m}$ but small heading ($\theta \approx -11°$ to $0°$) achieve failure probabilities near 0.42–0.55.
2) **The minimum-risk corridor is offset.** The minimum failure probability across the grid occurs near $\theta \approx -11°$ to $0°$ rather than at $\theta = 0°$ exactly. This offset is consistent with the estimated lateral bias $\mu_{p_y} \approx -0.028\,\text{m}$: the perception system systematically underestimates lateral position, causing the controller to over-correct in the positive direction. A small negative heading partially compensates for this bias.
3) **Credible intervals are widest at the decision boundary.** The 95% CI widths range from < 0.01 in regions where $\widehat{P}(\text{fail}) \approx 0$ or ≈ 1, to ≈ 0.2–0.25 in the transition zone near $\theta \in [-15°, +15°]$, where outcome is most sensitive to the realization of noise parameters.

[risk_heatmap.png]
Posterior mean $\widehat{P}(\text{fail}|x_0)$ over the (p_y, θ) grid (left) and 95% credible interval width (right).

Fig. 1. Posterior predictive failure risk over the initial state space.

[risk_1d_slices.png]
$\widehat{P}(\text{fail}|x_0)$ as a function of θ_0 (right panel) for five lateral positions $p_y \in \{-1.0, -0.6, 0.0, +0.6, +1.0\}$ m.

Fig. 2. One-dimensional slices through the posterior predictive risk surface.

[posterior_predictive_check.png]
Observed nonzero error histograms (red) vs. posterior predictive samples (blue/green) for each state dimension.

Fig. 3. Posterior predictive check for the Gaussian error model.

V. Analysis

A. Quantifying the Conservatism Gap

The core finding of this analysis is the significant discrepancy between the HJ-based "unsafe" classification and the actual posterior predictive risk. In the RoVer-CoRe benchmarks, the BRT is constructed based on the extreme values of the error bound $\mathcal{E}$. Every state $x_0 \in \mathcal{R}$ is unsafe under *some* admissible error sequence — but our results indicate that for many boundary states, the probability of encountering such an extreme sequence under the true error distribution is substantially less than one.

We define the conservatism gap $\Delta_\mathcal{C}(\tau)$ as the region of the state space that is formally unsafe according to the BRT but exhibits posterior predictive failure probability below a risk threshold τ:

$$\Delta_\mathcal{C}(\tau) = \{x_0 \in \mathcal{R} : \widehat{P}(\text{fail} \mid x_0) < \tau\} \tag{12}$$

Without access to the precomputed HJ value function in the current experimental setup (see Section VI), we use the empirical observation that states at the boundary of the evaluated grid (near $|p_y| = 1.0$ to 1.4 m, small θ) represent boundary-BRT states in the RoVer-CoRe literature [1]. For these states, $\widehat{P}(\text{fail})$ ranges from 0.43 to 0.62 at $\theta \approx -11°$ to $0°$, well below the worst-case implication of certain failure.

The practical consequence is significant: a risk-averse operator using only the BRT would exclude states near $p_y = +1.0$ m, $\theta \approx -11°$ from the operational envelope, yet these states exhibit failure probabilities below 45% under realistic noise. Depending on the application's acceptable risk threshold τ (e.g., 10%, 20%), a substantial fraction of boundary BRT states would be reclassified as operationally acceptable.

B. Impact of Systematic Perception Bias

The identification of a lateral bias $\mu_{p_y} \approx -0.028$ m through MCMC sampling provides a mechanistic explanation for the observed asymmetry in the failure probability surface. In a perfectly unbiased perception system, the risk profile would be approximately symmetric about $\theta = 0°$. Instead, our results show a clear asymmetry: the minimum-risk heading is shifted negative by approximately 5–10°.

This asymmetry arises from the coupling between the perception bias and the proportional controller. The controller computes $u_t = -k_{p_y}\hat{p}_{y,t} - k_\theta \hat{\theta}_t$ based on the perceived state $\hat{x}_t = x_t + e_t$. A consistent negative bias $\mu_{p_y} < 0$ means the controller perceives the rover as closer to the left boundary than it actually is, inducing a rightward correction. This effectively acts as a hidden safety buffer for states already near the left boundary ($p_y < 0$) and a hidden risk factor for states near the right boundary ($p_y > 0$) [1].

This finding underscores that bias and variance are not interchangeable in safety-critical contexts. A perception system with small variance but significant bias can be more dangerous than one with larger variance and near-zero bias, because the bias couples predictably to the controller's feedback law.

C. Uncertainty Propagation and Time-Variance

Analysis of the time-varying empirical standard deviations confirms that perception reliability degrades significantly over the 5-second simulation horizon. The standard deviation of the lateral error σ_{p_y} grows from approximately zero at $t = 0$ to ≈ 0.20 m at $t = 3$ s in the RoverDark benchmark. By incorporating this growth into the simulation via the scaling factor $\sigma_t^{(d)}$, we ensure that failure risk is not uniformly distributed over the time horizon; rather, the system enters a "high-risk regime" in the latter portion of each trajectory.

This heteroscedastic structure is a key feature that a stationary Gaussian model would fail to capture. The time-varying scaling represents a pragmatic compromise between the fully parametric time-varying model (which would require fitting a growth rate from limited trajectory data) and the stationary model (which would overestimate early-trajectory risk and underestimate late-trajectory risk).

VI. Conclusion

A. Summary

This paper presented a Bayesian framework for quantifying the conservatism gap between formal worst-case safety verification and probabilistic operational risk in perception-based autonomous systems. Using data from the RoVer-CoRe benchmarks [1], we demonstrated that:

1) MCMC-based Bayesian inference on perception errors is tractable and informative: 6,000 simulator samples are sufficient to characterize the error distribution with tight posteriors and excellent convergence diagnostics ($\hat{R} \leq 1.002$, bulk ESS $> 4{,}900$).
2) The perception system exhibits statistically significant systematic bias in all three state dimensions — bias that couples to the feedback controller and shifts the minimum-risk heading by 5–10° from the geometrically obvious choice of $\theta = 0°$.
3) Initial heading angle dominates initial lateral position as a determinant of failure risk under realistic noise: even at the corridor center, large heading angles produce near-certain failure.
4) The Gaussian model fits the continuous error component well ($\hat{R} \approx 1$, posterior predictive check passed), but structural zeros from initialization and light-reset events reveal that the full error process is not identically distributed across time.

B. Limitations and Caveats

Several limitations constrain the interpretation of these results.

Small trajectory dataset. Only 6 trajectories are available per benchmark, all from a specific set of initial conditions. The learned error distribution may not generalize to operating conditions far outside the training distribution.

Simplified controller approximation. The OLS-fitted proportional controller achieves $R^2 = 0.017$ on the recorded control inputs, indicating that the linear model captures very little of the MPC's behavioral variance. The failure probabilities estimated in Section IV are therefore conditional on this simplified control law, not the actual MPC policy. A more faithful approximation using a learned surrogate model or direct replay of recorded control sequences would strengthen the results.

Heuristic failure bound. The lateral boundary $|p_y| \geq 1.5\,\mathrm{m}$ is a conservative heuristic stand-in for the true HJ-defined failure set $\mathcal{F}$. Without loading the precomputed BRT value function, we cannot directly compare $\widehat{P}(\text{fail})$ to $\mathbf{1}[x_0 \in \mathcal{R}]$ for the same failure criterion.

Independence assumption. The model treats errors as independent across time, but the structural zeros and time-varying standard deviation both indicate temporal dependence. An autoregressive or state-space error model would be more faithful to the data-generating process.

C. Future Work

Several natural extensions follow from this work.

Zero-inflated mixture model. The structural zeros motivate modeling the error process as a mixture: $e_t \sim (1 - p_t)\,\delta_0 + p_t\,\mathcal{N}(\mu, \sigma^2)$, where p_t captures the probability of being in an active-uncertainty state and could be modeled as a function of time or light-activation state.

Time-varying Gaussian process model. A Gaussian process prior on $\mu(t)$ and $\sigma(t)$ would provide a principled non-parametric treatment of the non-stationarity observed in the error data.

Direct BRT comparison. Loading the precomputed HJ value function from the RoVer-CoRe pipeline would enable a direct, quantitative comparison of $\widehat{P}(\text{fail} \mid x_0)$ against the binary BRT classification $\mathbf{1}[V(x_0) < 0]$ for the same failure set, directly operationalizing the conservatism gap $\Delta_{\mathcal{C}}(\tau)$.

Chance-constrained safe set. The most actionable engineering output would be a probabilistic safe set $\mathcal{S}(\tau) = \{x_0 : \widehat{P}(\text{fail} \mid x_0) \leq \tau\}$ and a quantitative comparison of its volume against that of the BRT complement $\mathcal{R}^c$. This would directly quantify how much of the state space is recoverable by moving from worst-case to probabilistic analysis, providing actionable guidance for mission planning and controller design.

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